

SUMMARY TABLE: GENERAL SCIENCE GRADE 10

Sub-Strand 2.2: Chemical Families — Teacher Reference (Pre-filled)

SUMMARY TABLE: GENERAL SCIENCE GRADE 10	
Sub-Strand	Sub-Strand 2.2: Chemical Families
Driving Question	DRIVING QUESTION: Why does each element behave and get used the way it does, and how can the periodic table help us predict which element is right for which job in Kenyan daily life?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 The Sufuria, Wire, Mabati, Salt, Chlorine and Sodium-Lamp Puzzle	We observed that everyday Kenyan objects are made from different elements: aluminium sufuria, copper wire, iron mabati coated with zinc, salt (NaCl), chlorine-treated Mombasa water and sodium-lamp light on Thika Road. We also saw sodium float, fizz, dart about and melt in water, and turn the solution pink with phenolphthalein.	We learned that elements have very different properties, and some, like sodium, are dangerous on their own, yet each is useful and chosen for a job. We now have many questions about why an element suits a job, recorded on the DQB under Properties, Reactivity, Uses and Safety.	Our first explanation is that each element is used for the job that its properties suit, but we cannot yet explain why sodium is violent alone yet safe in salt, or why chlorine is poisonous as a gas yet safe in tiny amounts in water. We will use the periodic table and later investigations to answer these questions.
Lesson 2 Chemical Families: Sorting the First Twenty Elements by Position in the Periodic Table	We sorted the first twenty elements by position on the periodic table. Li, Na and K are in Group 1 (alkali metals); Be, Mg and Ca are in Group 2 (alkaline earth metals); F and Cl are in Group 17 (halogens); He, Ne and Ar are in Group 18 (noble gases). H, B, C, N, O, Al, Si, P and S are not in any of these four families. The transition metals, including Fe, Cu and Zn, sit in the central block starting from Period 4.	Elements are grouped into chemical families by their position in the periodic table. Groups are vertical columns and periods are horizontal rows. Elements in the same group have the same number of outer electrons (Group 1 = 1, Group 2 = 2, Group 17 = 7, Group 18 = 8, with He = 2), which links group number to similar behaviour.	Sodium (Na, Thika Road lamp and salt) is an alkali metal, chlorine (Cl, Mombasa water) is a halogen, aluminium (sufuria) is a metal not in the four families, and copper (wire), iron and zinc (mabati) are transition metals. So the elements in our puzzle come from different parts of the periodic table, and a family's position can help us predict behaviour. We still need evidence on their properties and reactivity to explain why each element suits its job.
Lesson 3 Physical Properties of the Families: Finding Patterns in Density, Melting Point, Hardness	We tested Mg, Al, Cu, Fe and Zn for appearance, hardness, density and conductivity, and studied a data table for Na, K, Li, Mg, Ca, Fe, Cu, Zn, Al, Pb, Cl ₂ , Ne and Ar. Na, K and Li are soft with	Elements in the same chemical family share similar physical properties, and families differ from each other. Alkali metals are soft, low in density and low melting. Transition metals are hard, dense, high	Properties help explain why an element suits a job: aluminium is light and conducts heat for the sufuria, copper conducts electricity excellently for wire, and iron is strong for the mabati base. Sodium is soft,

and Conductivity	densities below water (0.97, 0.86, 0.53 g/cm ³) and low melting points (98, 63, 181 C). Mg, Ca, Fe, Cu and Zn are denser and melt at higher temperatures. The metals conduct electricity; Cl ₂ , Ne and Ar do not.	melting and good conductors. Position in the periodic table helps us predict physical properties, though there are exceptions such as soft, dense lead.	melts at only 98 C and floats, which helps us understand why it behaves differently from the metals we use for pots and wires. We still need to explain why the families differ and why sodium reacts so strongly with water, which we will investigate in Lesson 4.
Lesson 4 Alkali Metals: Why One Outer Electron Makes Sodium Dangerous Alone but Safe in Salt	Lithium fizzes slowly in water, sodium fizzes fast, melts into a ball and darts about, and potassium reacts very fast with a lilac flame. All three float and the solution turns pink with phenolphthalein. Freshly cut sodium is shiny but quickly dulls in air. (Record what you saw and heard in your own words.)	Alkali metals (Group 1) have one outer electron (Li 2,1; Na 2,8,1; K 2,8,8,1). Reactivity increases down the group: Li < Na < K. They react with water to form an alkali (hydroxide) and hydrogen gas, and with oxygen in air. They must be stored under paraffin oil, handled only by the teacher, and any contact with skin or eyes needs immediate flushing with running water.	Alkali metals are very reactive because each atom has a single outer electron that is lost very easily. Down the group the outer electron is further from the nucleus, so it is lost more easily and reactivity increases. This is why sodium reacts violently with water ($2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$) and is dangerous alone, which links to the sufuria, wire, mabati, salt, chlorine and sodium-lamp puzzle. We still need to explain why sodium combined as salt is safe.
Lesson 5 Alkaline Earth Metals: Is Magnesium as Dangerous as Sodium?	Magnesium ribbon showed no visible reaction in cold water, gave steady bubbles of hydrogen gas in dilute HCl (pop test positive), and burned with a bright white flame in air leaving white MgO powder. Sodium, by comparison, fizzes fast in water and melts.	Alkaline earth metals (Group 2) have two outer electrons (Mg 2,8,2). Magnesium is less reactive than sodium. The reactivity order is K > Na > Li > Ca > Mg. Equations are balanced by adjusting coefficients so atoms are conserved.	Both sodium and magnesium react by losing outer electrons, but magnesium reacts less readily, shown by no cold-water reaction and slower reaction with acid. Its position in the periodic table lets us predict this: $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$ and $\text{Mg} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2$. So reactivity helps decide which element suits which job and how safely it can be handled.
Lesson 6 Halogens and Noble Gases: Why Chlorine Cleans Mombasa Water and Neon Lights the Sign	Record what you observed about chlorine (colour, reaction with sodium, product NaCl), the water treatment data (about 0.5 mg/L of chlorine, 10 mg for 20 L), and the data for neon and argon (colourless gases, 2,8 and 2,8,8, boiling points -246 C and -186 C, unreactive, non-conductors).	Write what you learned: halogens (Group 17) have 7 outer electrons and are reactive non-metals; chlorine kills the bacteria that cause cholera and typhoid at a tiny dose, but is toxic in large amounts; noble gases (Group 18) have full outer shells (He 2, Ne 2,8, Ar 2,8,8) and are unreactive; the balanced equation $2\text{Na} + \text{Cl}_2 \rightarrow 2\text{NaCl}$.	Explain, using outer electrons, why Mombasa tap water treated with chlorine is safe for Mama Wanjiru's ugali, why sodium and chlorine make safe table salt together, and why unreactive noble gases such as neon suit signs and road lighting. State how this helps answer the driving question: how the periodic table helps us match an element to a job.
Lesson 7 Transition Metals: Matching Properties to Jobs (Copper, Iron, Zinc, Lead and Aluminium)	We examined samples of Fe, Cu, Zn and Al and a data table for Fe, Cu, Zn, Pb and Al: density (Fe 7.87, Cu 8.96, Zn 7.14, Pb 11.34, Al 2.70 g/cm ³), melting point (Fe 1538, Cu 1085, Zn 420, Pb 327, Al 660 C) and conductivity (Cu excellent). We also researched their uses: wiring, mabati, galvanising, batteries and sufuria.	Transition metals such as Fe, Cu and Zn are generally strong, dense, good conductors with high melting points, and each use matches a specific property. Fe is strong and cheap, Cu is an excellent conductor and ductile, Zn coats iron and protects it by sacrificial protection, Al is light and resists corrosion, and Pb is very dense	Mama Wanjiru's sufuria is aluminium because it is light, conducts heat and resists corrosion; the kettle wire is copper because it conducts electricity excellently and can be drawn into wire; the mabati roof is iron for strength coated with zinc because zinc is more reactive and corrodes first. Elements are chosen for properties,

		but toxic so its use is being reduced.	balanced against trade-offs such as lead's toxicity, cost and availability.
Lesson 8 Putting It Together: Choosing the Right Element for the Job	We observed and reviewed evidence: sodium floats, fizzes and turns the solution pink in water (alkali NaOH forms); Al is light, Cu conducts very well, Zn coats iron, neon is unreactive; chlorine is a greenish-yellow gas used at about 0.5 mg/L in Mombasa water; sodium vapour gives the yellow Thika Road lamp light.	We learned that an element's properties depend on its outer electrons and so on its position in the periodic table: Group 1 has 1 outer electron and is very reactive, Group 2 has 2 and is less reactive, Group 17 has 7 and is reactive, Group 18 is full and unreactive. We learned that transition metals are strong, dense conductors, and that safety, cost and availability also affect which element is chosen.	We explained that sodium is violent alone because its single outer electron is lost very easily, but as NaCl it is stable and safe to eat; chlorine kills microbes at a tiny safe dose; aluminium suits the sufuria, copper suits wire, zinc protects iron on mabati and neon suits signs. The periodic table lets us predict family behaviour and match element to job while balancing safety, cost and availability.

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